

The four leakage smells

If you see one of these, the accuracy number on the slide is not real. Point at this poster.

why it is a leak

the fix

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```
train_test_split(  
    shuffle=True)
```

Test rows end up preceding training rows. The model is scored on hours whose future it has seen.

Split by TIME. Always.
There is no exception for 'just a quick check'.

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```
rolling(w,  
    center=True)
```

The window spans $[t-12, t+11]$ for $w = 24$ — it swallows the target hour and its neighbours.

`y.shift(1).rolling(w, center=False)`
The window must END before t .

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```
scaler.fit(  
    all_data)
```

The mean and standard deviation of the future are subtracted from the past.

Fit every scaler, encoder and imputer on the training slice only, inside the backtest loop.

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```
exog =  
    realised_temp
```

At 14:00 you do not have tomorrow's realised temperature. You have a forecast of it.

Use the FORECAST of the covariate, and let its error widen your interval honestly.

Every feature, every split and every scaler must be computable using only data that existed at the forecast origin.